

```
import System.IO

main = withSocketsDo $ do
  h <- connectTo "www.shakthimaan.com" (PortNumber 80)
  hSetBuffering h LineBuffering
  hPutStr h "GET / HTTP/1.1\nhost: www.shakthimaan.com\n\n"
  contents <- hGetContents h
  putStrLn contents
  hClose h
```

You can now compile and execute the above code, and it returns the *index.html* contents as shown below:

```
$ ghc --make get-network-uri.hs
[1 of 1] Compiling Main                (get-network-uri.hs, get-
network-uri.o )
Linking get-network-uri ...

$ ./get-network-uri
HTTP/1.1 200 OK
Date: Sun, 05 Apr 2015 01:37:19 GMT
Server: Apache
Last-Modified: Tue, 08 Jul 2014 04:01:16 GMT
Accept-Ranges: bytes
Content-Length: 4604
Content-Type: text/html
...
```

You can refer to the network-URI package documentation at <https://hackage.haskell.org/package/network-uri-2.6.0.1/docs/Network-URI.html> for more detailed information.

The *whois* Haskell package allows you to query for information about hosting servers and domain names. You can install the package on Ubuntu, for example, using the following command:

```
$ cabal install whois
```

The *serverFor* function returns a *whois* server that can be queried for more information regarding an IP or domain name. For example:

```
ghci> :m + Network.Whois

ghci> serverFor "shakthimaan.com"
Loading package array-0.4.0.1 ... linking ... done.
Loading package deepseq-1.3.0.1 ... linking ... done.
Loading package bytestring-0.10.0.2 ... linking ... done.
Loading package old-locale-1.0.0.5 ... linking ... done.
Loading package time-1.4.0.1 ... linking ... done.
Loading package unix-2.6.0.1 ... linking ... done.
Loading package network-2.6.0.2 ... linking ... done.
Loading package transformers-0.4.3.0 ... linking ... done.
Loading package mtl-2.2.1 ... linking ... done.
```

```
Loading package text-1.2.0.4 ... linking ... done.
Loading package parsec-3.1.9 ... linking ... done.
Loading package network-uri-2.6.0.1 ... linking ... done.
Loading package split-0.2.2 ... linking ... done.
Loading package whois-1.2.2 ... linking ... done.
```

```
Just (WhoisServer {hostname = "com.whois-servers.net", port =
43, query = "domain"})
```

You can use the above specific information with the *whoisI* function to make a DNS (domain name system) query:

```
ghci> whoisI "shakthimaan.com" WhoisServer {hostname = "com.
whois-servers.net", port = 43, query = "domain "}
Just "\nWhois Server Version 2.0\n\nDomain names in the .com
and .net domains can now be registered\n
...
```

You can also use the *whois* function to return information on the server as shown below:

```
ghci> whois "shakthimaan.com"
Just "\nWhois Server Version 2.0\n\nDomain names in the .com
and .net domains can now be registered\n
...
```

The type signatures of the *serverFor*, *whoisI* and *whois* functions are as follows:

```
ghc> :t serverFor
serverFor :: String -> Maybe WhoisServer

ghci> :t whoisI
whoisI :: String -> WhoisServer -> IO (Maybe String)

ghci> :t whois
whois :: String -> IO (Maybe String, Maybe String)
```

The *dns* package provides a number of useful functions to make DNS queries and handle the responses. You can install the same on Ubuntu, for example, using the following commands:

```
$ sudo apt-get install zlib1g-dev
$ cabal install dns
```

A simple example of finding the IP addresses for the *haskell.org* domain is shown below:

```
ghci> import Network.DNS.Lookup
ghci> import Network.DNS.Resolver

ghci> let hostname = Data.ByteString.Char8.pack "www.haskell.
org"
```